



Fig. 2: Connections to make Arduino Mega 2560 board as AVR programmer

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

too will start blinking, and after some time everything will stop. The boot loader has now been burnt into the new ATmega328P chip.

Loading boot loader into ATmega328. If the chip that you have purchased is ATmega328 (without a P), it makes all the difference. The programmer window will say that the chip signature is different – that is no reason to worry.

ATmega328 chip has an onboard 8MHz internal oscillator. That means, in absence of a quartz crystal oscillator, the chip can still be programmed and work at 8MHz speed. Unfortunately, this is not visible in the boards section, unless boards.txt file of Arduino is changed to make so. Replace `/usr/share/arduino/hardware/arduino/`

boards.txt file with the new boards.txt file that you have downloaded.

Close Arduino IDE, if it is open and then restart Arduino IDE. Now, go to Tools → Board and select ATmega328 (on a breadboard with 8MHz internal clock) or Arduino Uno.

Go to `/usr/share/arduino/hardware/tools/` and replace `avrdude.conf` file with the new `avrdude.conf` file that you have just downloaded.

Keep a back-up of the old file as it will be required again while programming the original chips. In fact, we are just making fool of the IDE software now by telling it to program the same ATmega328P (original) chip with only a different 'signature' line.

Alternatively, you can manually open `avrdude.conf` file, search for the block starting with `ATMEGA328P` and under it find the signature line `signature=0x1e 0x95 0x0f`. Replace this line with the new signature `signature=0x1e 0x95 0x14; #modified by Bera for ATmega328.`

`Avrdude.conf` is a very big file. The first time I used it, I did it at a wrong location under a similar ATmega description and signature line, and the result was frustrating for me. Therefore, I recommend that you simply replace the file with the new one and it will work for all versions of IDEs.

Now, close the IDE if it was opened and then reopen Arduino IDE. Repeat the process of burning the boot loader as mentioned above. To be on the safe side, remove capacitor C1 before uploading `avr_isp.ino` onto Arduino Mega 2560 board. Now, put back the capacitor again, select the board (Arduino Uno) and burn the boot loader into ATmega328 chip. It should work now.

Remember, capacitor C1 should al-

ways be connected on the programmer board while programming the target chip, else it will give errors. Once the boot loader is successfully loaded, you can use this as a new Arduino board.

What next? All the fun is awaited! Find Blink program from the examples in Arduino IDE, open it and compile it. The Blink program is listed below.

```
void setup() {
  pinMode(13, OUTPUT); // The LED
  connected to pin no 19
}

void loop() {
  digitalWrite(13, HIGH); // set the LED
  on pin no 19
  delay(1000); // wait for a second
  digitalWrite(13, LOW); // set the LED
  off on pin no 19
  delay(1000); // wait for a second
}
```

To upload the above code into ATmega328, go to File → Upload Using Programmer. (Remember this is not a common upload program. You are going to upload it using Arduino as a programmer.)

All LEDs on the programmer and the LED connected on pin 19 will start blinking for some time and then everything will stop. The LED connected to pin 19 is the LED connected to digital pin 13, which will start blinking now to show that you have successfully burnt the chip with Blink program.

Now, the code is uploaded into the target chip through Mega 2560 board. The LED at pin 19 will blink at the rate defined in the code. Now, you may disconnect Mega 2560 board from ATmega328 circuit and connect 5V supply at CON1.

This ATmega328 board can also be interfaced with sensors, LCD modules, 433MHz transmitter and receiver modules, rotary encoders, joy sticks and servos. So, it may be assumed that it is a normal board and will work like other Arduino boards. ●

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